14.3 Image Compression

Anybody who's attempted sending across files over the web would know that the bulkier the file, the tougher it gets. Exchanging or saving such files also may encounter difficulties due to their copious size. This is where the format of the data becomes crucial. Luckily for us, there's a whole lot of image-compression software available and universally acceptable in the world today.

File compression basically shrinks files by (a) coding information more efficiently and (b) eliminating redundancies or repetitions. So if, for example, a large part of an image is constituted by a single colour or charge value (like the sky or a grassland), the program reduces it to just s few bytes – some to record the colour and some to record the number of pixels that should be rendered in that colour.

Image compression could be 'lossy' or 'lossless'. Lossless ones (like TIFF) are great for medical photography or high-fidelity image scans but for all general purposes, lossy compression programs (such as JPEG) suit us fine. File compression is pretty effective with photographs because we don't notice the minor loss of detail that results due to the attempt to compress and reduce the bit rate. About 80% of the data in an image file is redundant and can be eliminated through compression. JPEG (Joint Photographic Experts Group) and GIF (Graphics Interchange Format) already use algorithms similar to those used by ZIP compression programs. Thus 'zipping' .jpg files is a bit counterproductive because the program must compress a file that is already compressed. Zipping works best with BMP images but nobody in the whole wide world uses bitmap files to store photographic images for the obvious reason that they're not compressed at all and therefore hog a lot of the disk space.

The most common image formats are TIFF, PSD, BMP, PCX, GIF AND JPEG, some of which we'll discuss here (you'll find the full monty on all these compression formats in Chapter 4 earlier).

TIFF: The Tagged Image File Format (now under the control of Adobe systems) is a versatile compressing program which can be compressed or uncompressed and can be stored